Cantera vs NN | Test IC 04 | NRMSE [%]

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	2.09	4.55	10.8	18	0.448	42.5	3.54	0.413	1.47	N/A	Complete
2 P-only DeepONet10	**10k**	1.69	0.88	3.02	1.68	0.186	0.792	0.222	0.573	0.937	N/A	Complete
2 P-only DeepONet10	**long**	0.394	0.15	0.279	0.469	0.138	2.82	0.239	0.176	0.28	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.91	1.07	3.57	1.99	0.204	1.58	0.292	0.43	0.705	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.142	0.0889	0.211	0.201	0.039	1.53	0.104	0.1	0.138	N/A	Complete
4 T/P DeepONet10	**10k**	3.65	0.816	4.14	3.02	0.403	11.1	0.928	0.931	0.984	N/A	Complete
4 T/P DeepONet10	**long**	29.7	7.02	30.3	40.7	0.323	161	11.1	1.51	370	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	3.08	0.662	3.39	2.06	0.378	10.8	0.842	0.646	0.637	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	21.4	4.5	17.5	21.7	0.14	163	10.4	0.323	2.07	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	13.8	3.79	17.2	6.25	0.921	49.8	2.22	5.91	83.6	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	22.5	7.15	31.8	14.9	0.464	37.6	2.84	1.02	7.26	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.908	0.267	1.08	0.408	0.166	3.2	0.265	0.445	0.374	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	19.6	6.75	28.7	10.3	0.305	37.6	2.45	0.801	6.42	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 05 | NRMSE [%]

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	43.9	43.5	2.94	11.3	54.9	501	42.3	11.2	14.7	N/A	Complete
2 P-only DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
2 P-only DeepONet10	**long**	88.2	80.3	5.2	92	73.9	22.7	81.8	61.5	14.2	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	39.8	42.2	51.6	9.03	41.6	53	40.7	10.7	14.6	N/A	Complete
4 T/P DeepONet10	**10k**	51.5	53	8.07	9.78	54	58	52	10.7	14.6	N/A	Complete
4 T/P DeepONet10	**long**	80.5	54.2	147	8.09	83.9	1.1e+03	55.6	10.9	14.6	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	50	51.4	5.78	8.03	52.4	58.8	50.4	9.98	14.5	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	55.1	50.2	46.5	10.7	73.2	825	47	9.69	14.5	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	62.7	49.4	188	6.52	58	575	45.8	83.3	14.1	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	44.8	49.8	84.1	8.55	56	370	43.3	22.6	14.5	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 07 | NRMSE [%]

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	30.9	26.8	45.6	1.64e+04	24.3	42.4	38.2	24.4	18.7	N/A	Complete
2 P-only DeepONet10	**10k**	38.8	40.7	45.6	31	42.2	41.9	38.1	45.7	40.3	N/A	Complete
2 P-only DeepONet10	**long**	1.54e+07	9.83e+06	1.17e+08	1.19e+10	4.83e+06	4.38e+10	2.74e+07	3.01e+03	3.15e+04	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	38.8	40.7	45.6	28.4	42.2	41.9	38.1	45.7	40.3	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.95e+07	1.25e+07	6.74e+07	1.50e+10	6.12e+06	5.36e+10	3.47e+07	5.4e+03	5.82e+04	N/A	Complete
4 T/P DeepONet10	**10k**	39	40.6	1.35e+03	45.3	42.2	66.4	38.1	45.7	40.3	N/A	Complete
4 T/P DeepONet10	**long**	4.20e+06	1.52e+07	2.85e+10	5.83e+09	9.49e+06	1.40e+12	2.07e+07	1.8e+03	3.57e+04	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	39.1	38.8	4.81e+03	44.3	39.3	599	37.8	39.6	39.7	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	5.60e+06	1.81e+06	3.13e+10	1.70e+07	5.16e+04	6.64e+08	1.75e+05	6.71e+03	8.43e+04	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	2.02e+07	1.45e+07	2.47e+09	1.38e+10	7.52e+06	2.83e+11	3.73e+07	4.91e+03	2.42e+04	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	103	26.2	4.44e+05	27.4	39.7	1.37e+04	30.5	43.5	40.3	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	1.89e+07	1.38e+07	2.39e+09	9.62e+09	7.25e+06	3.16e+11	3.50e+07	5.47e+03	3.63e+04	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	4.12e+05	2.61e+05	4.67e+03	2.97e+04	1.51e+05	1.23e+04	5.87e+05	8.1e+03	1.27e+05	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 18 | NRMSE [%]

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	4.16	6	11.4	5.57	3.77	85.4	6.67	4.69	4.07	N/A	Complete
2 P-only DeepONet10	**10k**	10.9	8.4	11.6	6.12	8.64	36	8.86	7.08	8.04	N/A	Complete
2 P-only DeepONet10	**long**	1.2	2.1	2.65	2.17	1.91	2.31	1.79	2.15	2.28	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	4.73	5.19	4.52	4.39	5.19	13	5.06	6.03	6.35	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.93	2.94	3.57	3.07	2.71	3.71	2.59	3.2	3.25	N/A	Complete
4 T/P DeepONet10	**10k**	10.6	8.84	15.3	6.18	8.75	45.3	8.23	9.08	19.3	N/A	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.552 ms**
5 ANN10 + T/P DeepONet10	**10k**	11.1	7.78	14.8	5.93	7.79	44.7	7.91	7.43	7.5	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	60.5	13.7	104	2.5	1.45	238	15.4	1.5	1.97	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	**long**	75.7	17.3	126	15.9	6.39	243	23.4	126	7.37e+04	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	49.7	9.18	49.4	3.65	9.37	374	26.3	9.37	19.7	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	63.5	11.7	103	7.32	0.634	258	17.9	1.14	1.02	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 32 | NRMSE [%]

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	5.24	5.76	23.7	3.74	0.329	39.4	3.17	0.324	2.22	N/A	Complete
2 P-only DeepONet10	**10k**	10.6	2.88	15.5	7.26	2.76	46.1	4.91	3.43	3.27	N/A	Complete
2 P-only DeepONet10	**long**	0.37	0.881	1.4	1.21	0.743	0.964	0.685	0.77	0.521	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.45	1.84	2.4	2.02	1.7	4.71	1.62	2.14	2.16	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.299	0.567	1.71	1.22	0.281	1.28	0.334	0.283	0.2	N/A	Complete
4 T/P DeepONet10	**10k**	6.72	5.81	15.5	7.33	5.21	47.1	3.46	5.54	28.1	N/A	Complete
4 T/P DeepONet10	**long**	77.3	19.8	156	6.73	1.25	230	17.4	2.23	66.9	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	9.03	1.91	15.4	6.68	1.13	46.5	2.91	2.45	2.15	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	7.23	1.07	10.4	1.78	0.86	41.9	3.14	0.971	0.779	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	48	4.14	50	9.62	2.13	320	25.3	7.18	15.9	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	118	22.1	211	15.8	0.877	406	31.8	78.3	1.17e+03	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	45.8	5.94	38.2	0.927	0.119	363	27.2	5.27	21.4	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	44.6	0.933	58.2	0.329	0.109	271	20.2	2.87	3.48	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 39 | NRMSE [%]

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	28.8	2.32	9.08	12.5	16.8	370	24.1	6.81	9.36	N/A	Complete
2 P-only DeepONet10	**10k**	0.755	1.15	3.32	16.4	0.341	21.7	0.661	2.89	2.68	N/A	Complete
2 P-only DeepONet10	**long**	0.0471	0.0723	0.267	0.179	0.17	1.8	0.0287	0.403	0.546	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	1.07	1.13	3.08	18.2	0.374	22.3	0.699	2.55	1.99	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	0.122	0.178	0.985	1.16	0.19	1.57	0.155	0.33	0.445	N/A	Complete
4 T/P DeepONet10	**10k**	7.79	1.03	3.04	9.45	3.31	60.4	4.88	6.71	8.88	N/A	Complete
4 T/P DeepONet10	**long**	4.79	0.203	1.62	1.18	2.68	51.9	3.4	1.79	2.53	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	6.96	0.633	2.49	7.87	3.52	60.1	4.36	7.28	9.86	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	65.1	1.85	50.5	6.09	29.8	634	41.9	7.95	10.8	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	7.83	0.959	3.45	7.86	3.48	64.4	4.98	5.54	6.34	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	0.0837	0.0861	0.123	0.248	0.0971	0.736	0.0874	0.149	0.175	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	6.74	0.59	3.05	4.33	3.34	63	4.46	5.99	7.74	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0724	0.244	0.405	1.66	0.453	7.13	0.176	0.198	0.881	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 44 | NRMSE [%]

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	36.2	24.8	9.96	14.2	25.2	370	34	9.57	8.29	N/A	Complete
2 P-only DeepONet10	**10k**	39.5	25.3	37.6	14.1	27.7	334	31.4	15.7	9.28	N/A	Complete
2 P-only DeepONet10	**long**	29.1	29.3	3.87	5.66	29.3	20.8	29.2	8.93	6.94	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	30.3	27.8	7.8	14.5	32.9	343	29.7	8.93	6.63	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	27.2	27.5	4.48	5.37	27.4	19.7	27.4	9.22	7.29	N/A	Complete
4 T/P DeepONet10	**10k**	26.5	28.2	19	6.91	28.6	43.2	27	93.4	5.51e+03	N/A	Complete
4 T/P DeepONet10	**long**	59	32.4	76.2	7.03	41.9	568	39.3	9.35	6.81	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	122	35	255	13.7	52.5	41.1	68.7	1.91e+03	2.52e+04	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	59	31.4	66.2	10.2	43	633	40.5	9.5	6.92	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	93.7	93.7	4.41	14.2	93.7	64.4	93.7	8.64	6.73	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	26.2	26.3	3.03	6.33	26.4	17.8	26.3	9.51	8.03	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	93.7	93.7	4.41	14.2	93.7	64.4	93.7	8.63	6.73	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	23.4	23.6	3.21	5.99	23.7	20	23.5	9.31	8.32	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs NN | Test IC 45 | NRMSE [%]

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	1.39	6.72	37.3	18.6	1.34	66.6	6.28	4.55	17.9	N/A	Complete
2 P-only DeepONet10	**10k**	42.2	28.1	17	96.9	13	176	36.4	5.65	23	N/A	Complete
2 P-only DeepONet10	**long**	34.4	25.1	18.1	96.1	12.7	114	30.1	0.39	1.84	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	25	17.4	16.9	34.5	7.44	142	23.5	6.96	40	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	1.76	3	7.42	5.03	1.69	10.2	2.81	1.23	1.43	N/A	Complete
4 T/P DeepONet10	**10k**	23.1	18.8	34.8	14.5	18.6	74.6	19.9	11.3	16.7	N/A	Complete
4 T/P DeepONet10	**long**	60.6	3.47	105	11.7	3.11	384	33.4	16.4	14.1	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	11.6	2.98	18.1	16.9	3.36	70.5	6.49	3.36	2.12	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	51	7.85	58.3	14.2	3.68	391	33.4	11.8	18.7	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	14.1	13.5	16.1	26.9	13.9	62.9	13.4	8.09	17.3	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	40.1	3.43	64.4	8.96	1.87	266	23.3	8.51	9.3	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	10	3.21	18.5	16.8	4.53	70.5	4.99	2.52	1.92	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	1.18	0.546	1.56	0.834	0.464	9.47	0.898	0.383	0.923	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.